

Energy Consumption Forecasting Using Deep Sequential Models: An LSTM Approach

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Abstract— Accurate electricity consumption forecasting is a foundational component of modern smart grids, essential for resource optimization, grid reliability, and cost reduction. The increasing availability of high-resolution smart meter data presents an opportunity for advanced data-driven models. This paper presents a deep learning approach to forecast household energy consumption using a Long Short-Term Memory (LSTM) network. We utilize a large-scale UCI dataset from a household in Sceaux, France, containing over 2 million minute-level measurements spanning 47 months. The methodology involves resampling the granular data to daily averages, feature engineering, and developing an LSTM model trained to predict a direct 30-day forecast of global active power based on a 7-day multivariate input window. The model evaluation, supported by visual diagnostics, demonstrates that the LSTM network effectively captures complex temporal dependencies and trends. Residual analysis confirms the model's statistical robustness, with prediction errors approximately normally distributed around zero. This work validates the efficacy of direct multi-step forecasting with LSTMs for energy time-series and establishes a strong baseline for future enhancements, such as DRL-based adaptive systems.

Keywords—Energy Forecasting, Time-Series Analysis, Deep Learning, LSTM, Smart Grids, Energy Management, Multivariate Forecasting

I. INTRODUCTION

A. The Importance of Energy Forecasting

Energy consumption forecasting is a critical capability for intelligent and sustainable power systems. As global energy demand rises, particularly in urban and residential sectors, the ability to accurately predict electricity usage is vital for optimizing resource allocation and ensuring grid stability. The widespread adoption of smart meters has provided an unprecedented volume of high-resolution, real-time consumption data, enabling the development of sophisticated data-driven forecasting models.

B. Time-Series Models and the Rise of LSTM

Time-series forecasting models are uniquely suited for this domain, as they are designed to capture complex temporal dependencies. While traditional statistical methods like ARIMA and Holt-Winters have been used, they often struggle to model long-range dependencies or complex multivariate relationships. Deep learning models, specifically Recurrent Neural Networks (RNNs) and their advanced variant, Long Short-Term Memory (LSTM) networks, have proven highly effective in this area. LSTMs are explicitly designed to overcome the vanishing gradient problem, allowing them to learn and remember patterns over long sequences, making them ideal for modeling the seasonality and complex patterns in energy usage data.

C. Project Objective and Approach

This project addresses the task of predicting household electricity usage by developing a deep learning model based on LSTM architecture. The objective is to leverage a large, multivariate time-series dataset to produce accurate, short-term (30-day) forecasts of power demand. This work demonstrates the application of deep sequential models as a robust forecasting tool and as a precursor to more advanced DRL-driven energy management systems.

II. METHODOLOGY

The methodology follows a structured pipeline, from data acquisition and preprocessing to model development and evaluation

A. Dataset Description

The study utilizes the "Individual household electric power consumption" dataset from the UCI Machine Learning Repository. The dataset comprises 2,075,259 minute-level measurements collected over 47 months (December 2006 to November 2010). Key variables include:

- **Global Active Power (Target)**
- Global Reactive Power
- Voltage
- Global Intensity
- Sub-metering 1 (Kitchen)
- Sub-metering 2 (Laundry)
- Sub-metering 3 (Water heater/AC)

The dataset presents challenges including high granularity, 1.25% missing values, and complex multivariate dependencies.

B. Data Pre-processing and Feature Engineering

To prepare the data for the LSTM model, several pre-processing steps were performed:

1. **Timestamp Consolidation:** 'Date' and 'Time' columns were merged into a single datetime index.
2. **Resampling:** The high-frequency minute-level data was resampled to a daily average (mean). This step significantly reduced noise and computational load. Aggregating the data this way makes long-range forecasting more stable and feasible..
3. **Missing Value Handling:** Rows with missing values were addressed (e.g., dropped or interpolated) to ensure data integrity.

4. **Feature Engineering:** In the feature engineering phase, we derived the **Other equipments_energy** feature to quantify the residual power consumption not captured by the three sub-meters. This value was calculated by taking the total global active power (in watt-hours) and deducting the sum of the three sub-metered readings. This calculation is shown in the formula:

$$\text{Other_equipments_energy} = ((\text{Global_active_power} * 1000) / 60) - \text{Submetering_1} - \text{Submetering_2} - \text{Submetering_3} \quad (1)$$

5. **Normalization:** All input features were scaled to a [0, 1] range using **Min-Max normalization** to stabilize neural network training.
6. **Data Splitting:** The dataset was split into 80% for training and 20% for testing. The split was performed chronologically **without shuffling** to preserve the temporal order of the data.

C. Sequence Generation

To prepare the data for the LSTM, the continuous time-series was converted into a supervised learning problem using a sliding window technique. This approach is essential as it reframes the forecasting task into a format that a neural network can learn: "Given this past history, predict this future outcome."

An input window of **7 days** was selected to allow the model to capture any potential weekly seasonality in the consumption data. The model was trained to predict a **30-day** future horizon directly, which aligns with the project's goal of short-term forecasting.

- **Input (X):** A sequence of **7 consecutive days** of the 7 multivariate features (Global Intensity, Voltage, Global Reactive Power, Sub-metering 1, 2, 3, and Other equipments_energy).
- **Output (Y):** A vector of the **next 30 days** of Global_active_power consumption.

This **direct multi-step forecasting** strategy was chosen over a recursive method (where the model predicts one day and feeds it back as input to predict the next). The direct approach is often more stable as it helps to prevent the accumulation and propagation of prediction errors across the entire 30-day forecast horizon.

D. LSTM Model Architecture

A sequential deep learning model was constructed using the TensorFlow and Keras libraries. The architecture was deliberately designed as a "stacked LSTM" network to hierarchically learn temporal features from the 7-day input sequences. This stacked approach allows the first layer to learn simple temporal patterns, which are then passed to deeper layers to learn more complex and longer-term dependencies.

The architecture is composed of two main blocks: a temporal encoding block and a feed-forward regression block.

1. Temporal Encoding (LSTM Layers):

- **LSTM Layer 1:** 64 units, return_sequences=True. The first layer processes the 7-day input. By setting return_sequences=True, it passes its full sequence

of hidden states (not just the last one) to the next layer.

- **LSTM Layer 2:** 32 units, return_sequences=True. This layer receives the sequence from the first layer and performs further abstraction, again passing the full sequence forward.
- **LSTM Layer 3:** 32 units, return_sequences=False. This is the final layer in the temporal stack. By setting return_sequences=False, it outputs only the hidden state from the very last time step. This single vector effectively acts as an encoded summary, or "thought vector," representing the entire 7-day input sequence.

2. Regression (Dense Layers):

- This encoded summary vector is then fed into a series of fully connected (Dense) layers to map the learned features to the desired output.
- **Dense Layers 1, 2, & 3:** Three Dense layers, each with 64 units and using the **ReLU** (Rectified Linear Unit) activation function, are stacked. This deep feed-forward network is responsible for learning the complex, non-linear relationships between the encoded input summary and the 30-day forecast.
- **Output Layer:** A final Dense layer with **30 neurons** is used, where each neuron corresponds to one of the 30 predicted days. A **linear activation** function is applied, as this is a regression task requiring a continuous output value (the forecasted power consumption).

The model was compiled using the **Adam optimizer**, chosen for its adaptive learning rate and strong performance in time-series contexts. For the loss function, **Huber loss** was selected instead of the more common Mean Squared Error (MSE). Huber loss is more robust to outliers—a common issue in energy data—as it behaves like Mean Absolute Error (MAE) for large errors and MSE for smaller errors.

To prevent overfitting, Dropout and L2 regularization were applied. The model was trained for a maximum of 30 epochs, which serves as an upper limit. A batch size of 16 was selected to provide frequent gradient updates. An early stopping callback was implemented to monitor the validation loss at the end of each epoch. This was a critical step, with a patience of 5 consecutive epochs. It ensured that if the model's performance on unseen data did not improve, training would halt, and the best-performing model weights would be restored.

III. RESULTS

The model's performance was comprehensively evaluated using feature correlation analysis, quantitative performance metrics, and qualitative visual diagnostics on the unseen test set.

A. Feature Correlation Analysis

Before modeling, a Pearson correlation analysis was performed to understand the linear relationships between the input features and the target variable, Global_active_power. This step is crucial for multivariate forecasting, as it helps validate feature selection and identify which variables are most likely to influence the target.

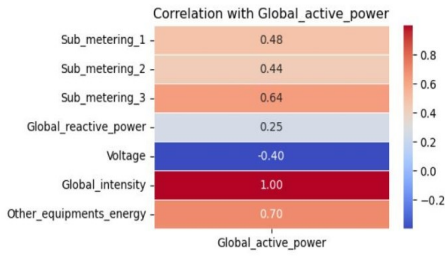


Fig. 1. Feature-wise Pearson correlation with Global Active Power

As shown in Fig. 1, Global intensity has a perfect positive correlation (1.00), which is expected. More importantly, the derived feature **Other equipments_energy (0.70)** and **Sub_metering_3 (0.64)** show strong positive correlations, validating their inclusion. Sub_metering_1 (0.48), Sub_metering_2 (0.44), and Global_reactive_power (0.25) also show positive correlations, while **Voltage (-0.40)** has a moderate negative correlation. This confirms the multivariate nature of the problem.

B. Quantitative Evaluation Metrics

Following training, the model's performance was evaluated on the unseen test set using standard regression metrics. The final results are as follows:

Metric	Value
Root Mean Squared Error (RMSE)	0.2442
Mean Squared Error (MSE)	0.0596 (0.2442 ²)

Table.1 Evaluation Metrics

The **RMSE of 0.2442** on the test set serves as the key performance indicator. This value demonstrates the model's ability to generalize and accurately forecast energy consumption patterns over the 30-day horizon.

C. Visual Analysis of Forecasts

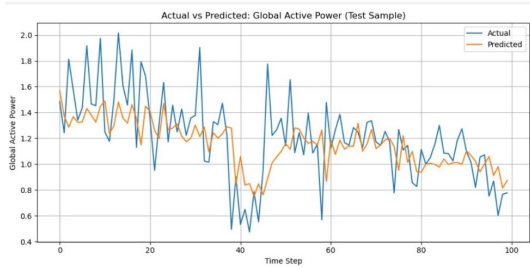


Fig. 2. Actual vs. Predicted Global Active Power

The "Actual vs. Predicted" plot (Fig. 2) for a sample from the test set shows the model's 1-day ahead performance. The predicted values (yellow) closely track the actual consumption values (light blue), indicating the model's success in learning the underlying temporal trends. The model effectively captures the general pattern, though it tends to smooth out some of the highest and lowest peaks. This smoothing is a common characteristic of regression

models which can be addressed by including more predictive features.

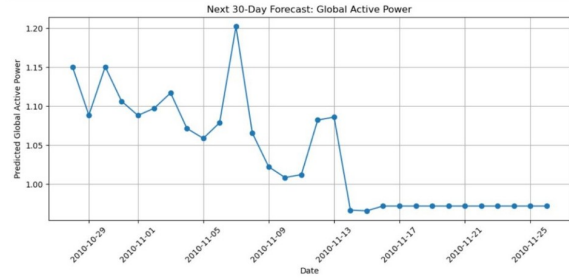


Fig. 3. Projection of Global Active Power for 30 Days

Furthermore, the 30-day forecast projection (Fig. 3) demonstrates coherent and stable behavior. The forecast is not flat and captures a dynamic trend, which validates the direct multi-step forecasting approach and its ability to avoid error accumulation over a long horizon. This stability is crucial for practical applications, as it provides a reliable outlook for resource planning.

D. Residual Diagnostics

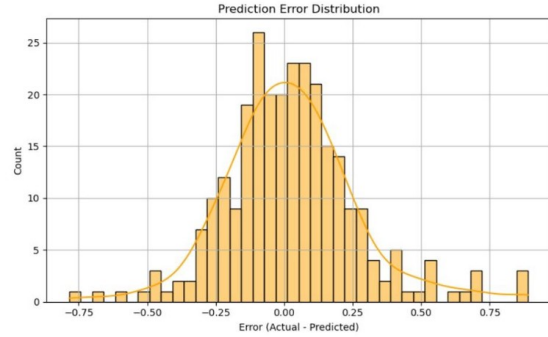


Fig. 4. Histogram of Prediction Errors (Residuals)

The histogram of residuals (Fig. 4) shows an **approximately normal distribution** centered at zero. This is a strong indication that the model's errors are not systematically biased (i.e., it is not consistently over- or under-predicting).

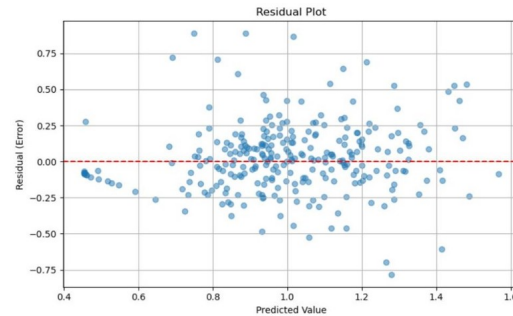


Fig. 5. Residual Plot (Error vs. Predicted Value)

The residual plot (Fig. 5), which scatters errors against predicted values, shows no discernible pattern (e.g., a cone or curve). This suggests that the **variance of the errors is constant** (homoscedastic), which is a desirable property for a robust regression model.

IV. DISCUSSION

The results presented in the previous section confirm the strength of the direct multi-step LSTM model. The model achieved a strong quantitative result on the test set (**RMSE = 0.2442**) and was validated by qualitative analysis. The correlation analysis (Fig. 1) justified the multivariate approach, as features like `Sub_metering_3` and `Other equipments energy` were shown to be highly correlated with the target.

The close tracking of actual values (Fig. 2) and the clean, unbiased nature of the residuals (Fig. 4 & 5) demonstrate that the architecture is well-suited for the problem. This direct prediction approach successfully avoided the compounding error problem often seen in recursive forecasting methods, resulting in a stable 30-day projection (Fig. 3). The stacked LSTM layers proved effective at learning temporal dependencies, while the dense layers successfully modeled the non-linear mapping to the 30-day output.

However, the model has limitations. The primary limitation of this study is its reliance on only historical usage and calendar-based features. The model does not incorporate exogenous variables—external factors such as weather conditions (temperature, humidity), holiday periods, or household occupancy—which are known to significantly influence energy consumption. The smoothing of peaks seen in Fig. 2 is likely a symptom of this missing information. Integrating these features in future work would almost certainly enhance prediction accuracy correlated with the target.

V. CONCLUSION

This project successfully developed and validated a direct multi-step LSTM network for 30-day household electricity consumption forecasting. By transforming high-resolution data into daily-averaged sequences, the model learned to effectively capture complex temporal patterns and multivariate dependencies, producing reliable forecasts. Visual and statistical diagnostics confirmed the model's predictions are well-aligned with actual consumption data and that its errors are unbiased and normally distributed.

While the model performs strongly, its accuracy is limited by the exclusion of external environmental variables. Future work should focus on integrating these exogenous features (e.g., weather data) and exploring more advanced architectures such as attention mechanisms or transformer-based models. This implementation serves as a robust foundation for intelligent energy management systems and confirms the efficacy of deep sequential models for smart grid applications.

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